AR Hardware - Software

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AR Hardware

1. Sensors
2. Processors
3. Displays

Sensors

- Sensors acquire information about the real world and communicate that information to the AR application.
- Role of sensor is to provide information about the real world to the AR application for a variety of purposes (location and orientation of the participant or environment information) to the AR application.

Tracking

1

Optical Tracking

- Camera (gathers light through a lens and provides a signal that represents an image of what the camera “sees)

2

Acoustical Tracking

- Microphones (use acoustical information/ ultrasound to be sensed)

3

Electromagnetic Tracking

- Electromagnetic tracking systems are able to track in six degrees of freedom by using transmitter

4

Mechanical Tracking

- attaching linkages to the object you wish to track

5

Depth Sensor for Tracking

- to cover a number of different technologies that can be used to provide information

6

Multiple Sensor for Tracking

Optical Tracking

- optical tracking systems can be implemented by placing cameras in the environment and having them “watch” the entities being tracked or attaching them to the tracked entities and “watching” the environment
- Optical tracking is also used for motion capture systems used to capture the performance of complex motions of multiple entities.



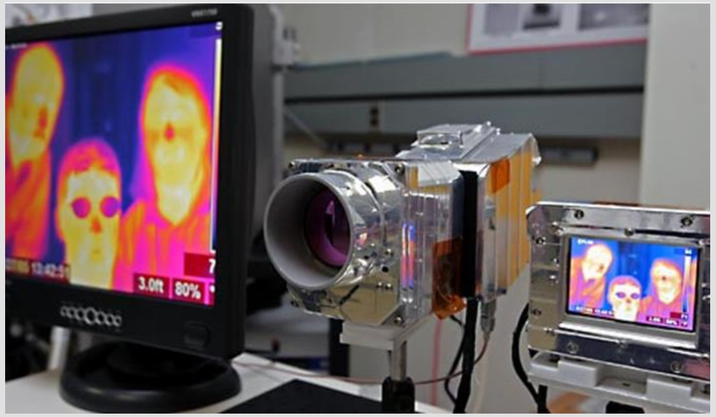
Smartphone camera – common use optical tracking



The Wii remote uses both optical tracking and an accelerometer to control the Nintendo Wii.
Photo courtesy of Holly McClain



small infrared camera can be used as a tracking sensor in AR systems. Note that there are LEDs surrounding the camera to bathe the scene with infrared light.
Photo courtesy of Liz Wuerffel



infrared camera developed by NASA “sees” light in a different frequency range than conventional cameras. This enables the camera to detect things in the dark or that are otherwise not visible to cameras that work in the visible spectrum.
Photo courtesy of NASA.

Optical Tracking

Pic. 1: Sensor on Participants (User)

- there is a single/ multiple cameras on the participant's head.
- having a sensor(s) on the participant that watches the environment from the perspective of the participant
- In some cases, such as mobile augmented reality applications on smartphones, the participant might be carrying the sensor(s) anyway



Pic.2: Sensor on environments

- there are single/ multiple cameras in the environment
- the environment watching what is happening in the environment and watching the participant(s)
- It may not be possible to place sensors in the environment, in that case—they must be connected in some way with the participant



Optical Tracking

Advantages:

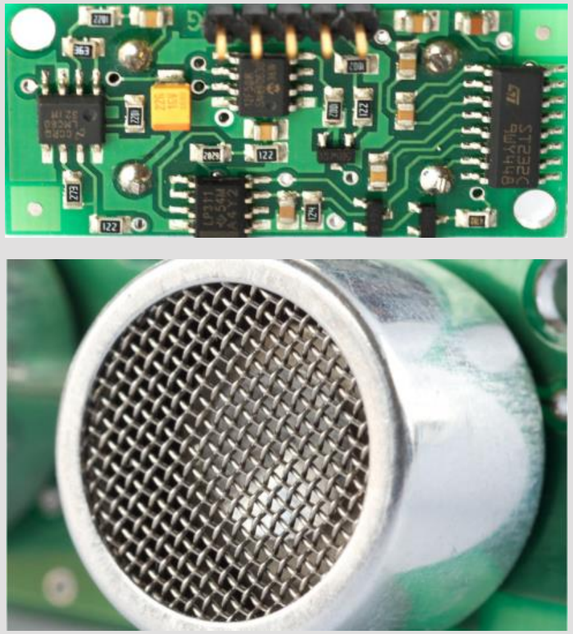
- can be done without requiring wires or anything else to be attached to the object being tracked
- allows numerous entities to be tracked simultaneously
- Small, relatively high-quality cameras become ubiquitous and thus are available very inexpensively
- doesn't require any alterations to the real world

Disadvantages:

- requires enough ambient light to "see" the real world in enough detail to provide suitable images to analyze by the computer vision software
- needs to be a clear "line of sight" between the camera and the real world and the entities being tracked
- the environment it is used in must provide information suitable for tracking
- Optical tracking systems introduce latency (takes time to acquire an image, deliver that image to the processor, and then analyze that image to determine the desired information)

Acoustical Tracking

- there needs to be a source of acoustical information to be sensed by the microphones (ultrasound).
- Ultrasound is sound that is higher in frequency than the human hearing system can perceive
- ultrasonic tracking systems work by having the object being tracked emit sound, and an array of microphones in the environment to capture that sound
- Based on the timing and amplitude of sound sensed in each microphone, the location of the source of the sound can be computed.
- objects can be tracked by having microphones attached to the objects and sound sources in the environment.
  - In this case, the objects being tracked would need to be connected to the computing system in order to report the signals their microphones are sensing



An ultrasonic sensor can be used to do acoustical tracking in an AR system. This photo pair shows both the front and the back of the sensor.
Photos courtesy of Liz Wuerffel.

Acoustical Tracking

Advantages:

- not being affected by lighting conditions

Disadvantages:

- cannot be used in environments that are noisy with audio signals in the same frequency range that the system uses
- each object to be tracked must have a sound source attached to it.
  - it can only be used in environments in which it is known a priori what objects will be tracked
- have limited range.

Electromagnetic Tracking

- Electromagnetic tracking systems are able to track in six degrees of freedom.
- implemented is that there is a transmitter with three orthogonal antennas.
- There is a corresponding receiver attached to the entity you wish to track that also has three orthogonal antennas.
- The sensor is the receiving unit.
  - The transmitter emits a signal, sequentially through each of its antennas. The signal acquired by each of the antennas is then analyzed, and the level of signal reported by each antenna can be used to compute the location and orientation of the receiver



This is the transmitting unit of an electromagnetic tracking system. The transmitter is in a fixed position, and the receiving units (small objects about the size of a die) are attached to the object being tracked. There are different sizes of transmitting units. This one serves a radius of about 12 feet.
Photo courtesy of Hank Kaczmarski.

Electromagnetic Tracking

Advantages:

- very precise and accurate.
- not dependent on ambient light levels and, as such, can be used in completely dark environments, very bright environments, or anything in between.

Disadvantages:

- sensitive to metal in the environment that they are in.
- limited in range, there are limits to the usable range. In general, the receivers need to be within a few meters of the nearest transmitter
- costly

Mechanical Tracking

- Mechanical tracking operates by attaching linkages to the object to track.
- Those linkages have sensors at each of the joints that report the angle between the linkages.
  - This is done by placing a variable resistor (potentiometer) at the joint and reading the voltage there.
  - As the angle of the linkage changes, the amount of resistance in the potentiometer changes and a corresponding change in voltage (that can be measured) occurs.
- The voltage can be used to determine the angle between linkages.
- This information, in combination with the angles between all other linkages in the system, can be used to compute the location and pose of the object



mechanical tracking can be done by measuring the angles between articulated joints. The primary use for this particular arm is for digitizing objects. It does this by measuring where the tip of the arm is and reporting that location for each point on an object that you touch with the tip. It can also be used to track any object by attaching the object to the tip of the arm.
Photo courtesy of Hank Kaczmarski

Mechanical Tracking

Advantages:

- fast and very precise.

Disadvantages:

- not possible to use mechanical tracking because the required mechanical connections can’t be used.

Depth Tracking

- can be used for tracking in augmented reality applications.
- The term depth sensor is used to cover a number of different technologies that can be used to provide information about how far away an object is from the sensor.
- The result from a depth sensor is a measurement of how far an object is from the depth sensor.
- This information can be used in conjunction with other tracking technologies to provide information about the location of an object.

Depth Tracking

Advantages:

- Depth sensors are relatively inexpensive.
- Depth sensors do not require anything to be attached to the objects being tracked.
  - there is no need to know ahead of time what objects need to be tracked.

Disadvantages:

- Depth sensors only report, as depth information.
- don't supply location or orientation information directly
- don't report what object they are "seeing", merely report the distance they are from whatever object is nearest to them

Multiple Sensor Tracking

- Each sensor can contribute to the overall goal of tracking.
- For example, accelerometers can be used to obtain information about relative motion but don't provide any information about an exact location.
- Additionally, many tracking systems that use accelerometers as their basis have a problem with error propagation the Wii system uses a combination of optical tracking and accelerometers
- the Wii system uses a combination of optical tracking and accelerometers
- The Xbox Kinect uses a combination of a depth sensor, optical tracking, and an array of microphones (acoustical tracking) to achieve a remarkable system that tracks the actions of a participant with no encumbering devices



The Microsoft Kinect uses multiple sensors, including a depth sensor to track the participant.
Photo courtesy of Michael Simeone



Depth Tracking

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Processor

- The core roles of the processor are **to receive the signals from the sensors**, **execute the instructions from the application** program based on the sensor information, and **create the signals** that drive the **display(s) of the system**
- The processing system in AR applications consists primarily of one or more general-purpose microprocessors as the **central processing unit (CPU)** and perhaps one or more special-purpose **graphics processing units (GPUs)**

Processors System Architecture

1. Application run on handheld system such as a smartphone
2. Application run on handheld system connected to remote server(s)
3. Application run on desktop/laptop comp
4. Application run on desktop/laptop computer connected to remote server(s)
5. Application run as a web application
6. Application run on a cloud with a thin client
7. Other combinations of local and remote systems

Capabilities of Different Processor System Architecture

Table 3.1 This Table Indicates the Capabilities of Several Different Architectures Available for Augmented Reality Systems

	Handheld	Handheld with Server	Desktop/ Laptop	Desktop/ Laptop with Server	Web Based	Cloud Based	Combination
Portable	Yes	Yes	No	No	No	No	Maybe
Portable with network	Yes	Yes	No	No	Yes	Yes	Maybe
Sufficient computation alone	Maybe	Yes	Yes	Yes	Yes	Yes	Maybe
Requires network	No	Yes	No	Yes	Yes	Yes	Maybe
Cross-platform solution	No	No	No	No	Maybe	Maybe	Maybe
Latency important	Maybe	Yes	No	Yes	Yes	Yes	Maybe

Displays

1 Visual Displays

- Stationary visual displays
- Visual displays that move with the participant's head
- Visual displays that move with the participant's hand or other parts of body

2 Audio Displays

- Stationary audio displays
- Audio displays that move with the participant's head
- Audio displays that move with the participant's hand or other parts of body

3 Haptic Displays

- Skin sensations (taction)
- Forces (kinesthetics)

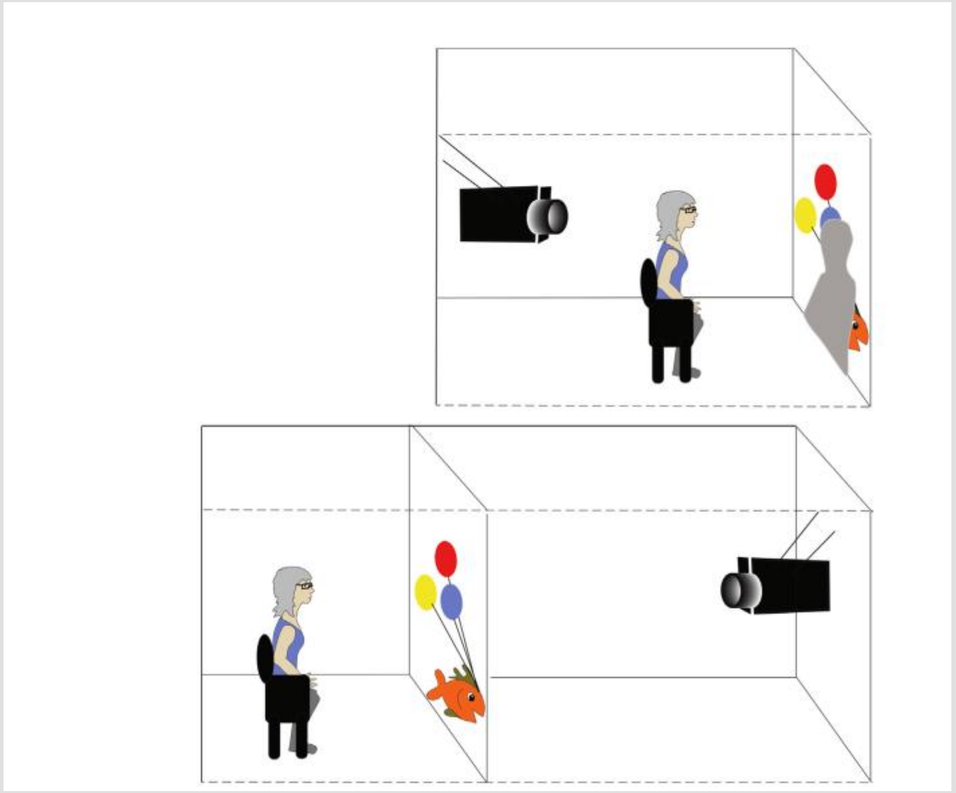
4 Smell

Stasionary Visual Display

- displays that don't move during their typical use.
- They are placed in position, and they remain there.



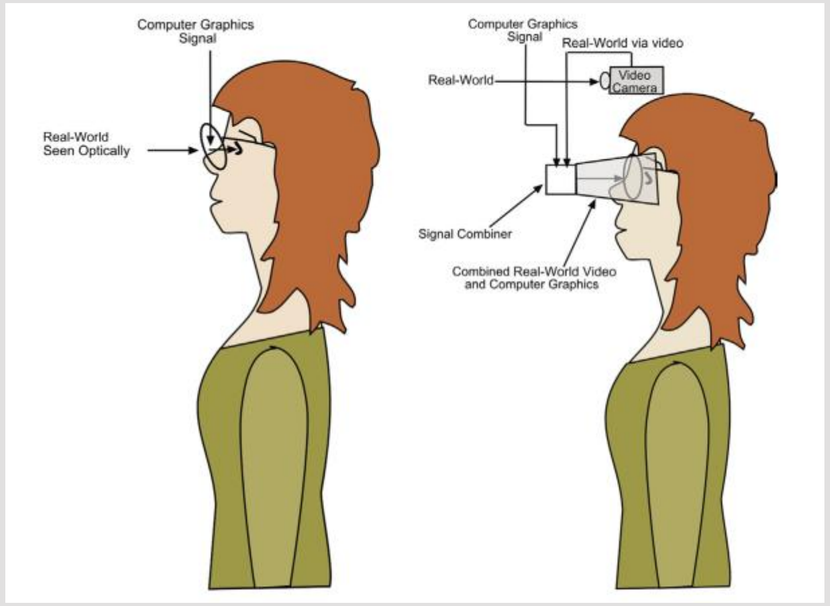
augmented reality is used in a retail setting. This particular installation is considered a kiosk and uses a stationary AR display. The participant holds up the product (in this case a Lego box) and can see an enhanced image of the box, as well as the box, on the display. The participant can manipulate the box to see the model from different perspectives. Note the fixed position camera above the display



Projection-based augmented reality - In AR environments, however, where the participant might be moving about in the room, shadows are caused when the participant passes between the projector and the surface on which it is projecting

Visual Display that Move with Participants Head

- Ex: HMD, helmets, glasses,headphones (audio)



- The first diagram illustrates the optical see-through type. In this type, the participant views the (real) world directly or perhaps through an optical lens.
- The computer graphics signal is then fed to the participant's eye(s) in conjunction with the optical signal, often by the use of a see-through mirror.
- The video see-through method illustrated in the second diagram generally precludes the participant from seeing the (real) world directly and instead provides a video camera (or potentially two video cameras for stereoscopic viewing) that is aligned as closely as possible with the participant's eye(s).
- This provides a video signal of the real world that can be combined with the video signal from the computer graphics system, which can then be displayed to the participant's eye(s)

Visual Display that Move with Participants Head

- Ex: HMD, helmets, glasses,headphones (audio)



Helmet-style, video see-through AR display. Image courtesy of Alan B. Craig



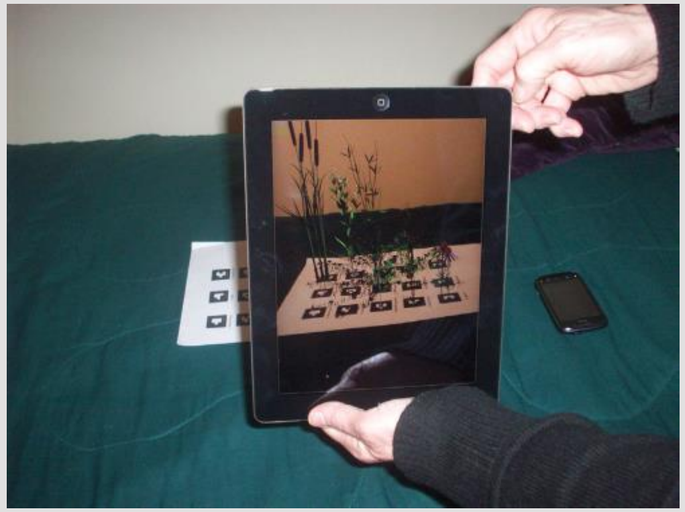
The nVisor ST50 from NVIS Inc. Photo courtesy of NVIS Inc



AR visual display as part of contact lenses. By using a lens in each eye, a stereoscopic display can be made

Visual Display that Move with The Participants Hands or other part of body

- The use of smartphones or tablet



Smartphones and smart tablets can be used as a “magic lens” to allow the participant to see things invisible without it. Here, a sheet of fiducial markers shows a garden of different plants when viewed through an iPad. Photo courtesy of Alan B. Craig

Visual Display that Move with The Participants Hands or other part of body

- The Sandbox

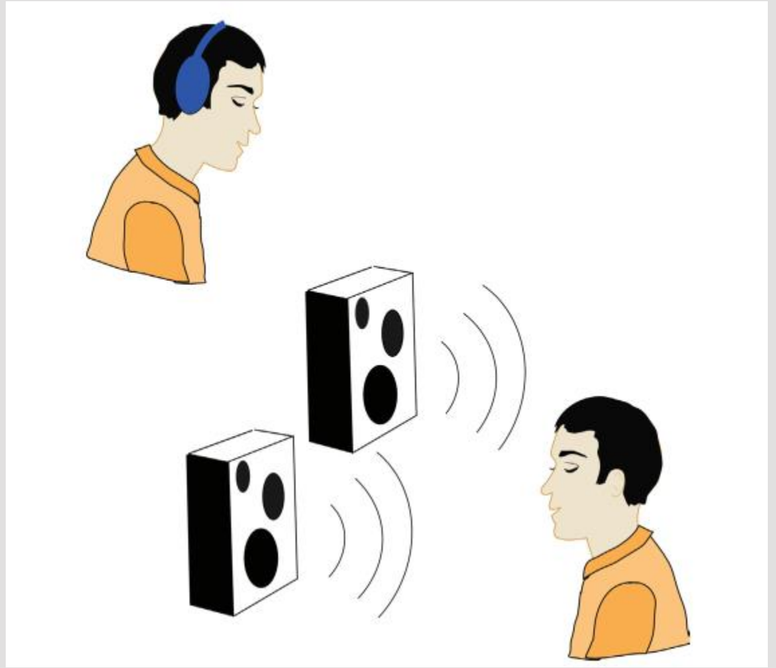




Stasionary Audio Display



front and rear views of a single high-quality speaker cabinet. In this case, the amplifier is contained in the same cabinet as the speakers, and the controls for the amplifier are on the rear panel.



This diagram shows the difference between using stereo headphones versus stereophonic speaker cabinets to deliver stereo audio

Audio Display that Move with Participants Head

- Headphones, earbuds



Earbuds and headphones both allow for private listening. Earbuds (left) are worn in the ear and are fairly unobtrusive visually. Headphones (right) are worn over the ears.
Photo courtesy of Olivia Neff

Audio Display that Move with The Participants Hands or other part of body

- Smartphones, smart tablets, and other handheld devices often have an audio display built in

Haptics

- Haptics, in general, refer to our sense of touch
- Skin sensations include things such as temperature, texture, and pain.
- Forces are sensed by how our body responds to them.



Skin sensations can be simulated with transducers that participants wear on their fingers. In this example (haptic canvas: dilatant fluid-based haptic interaction from the Bioimaging Laboratory, Osaka University), participants can “feel” sensations due to the transducers they wear on their hand and the fluid they interact with



This device from Geomagic is the Sensable PHANTOM Omni haptic device. It is a sensor that reports six degrees of freedom (X, Y, Z, yaw, pitch, and roll) as an input device (thus it is tracking your hand) and three degrees of freedom (X, Y, and Z) of output as force feedback. Photo courtesy of Geomagic. © Copyright Geomagic



Smell

- Smell is generally administered globally as a mist from an atomizer under computer control or more specifically to a particular person via tubing
- One thing that makes the use of smell problematic is that it is difficult to “clear” a smell rapidly when it should go away



- In this augmented reality application from Takuji Narumi of the University of Tokyo, a participant wears a see-through head-mounted display equipped with rubber tubing through which different scents can be presented to the participant’s nose.
- In this example, the participant tries a bite of a tasteless cookie on which there is an imprinted fiducial symbol.
- With the AR system, the participant sees the cookie as a specific type of cookie and is presented with the smell of that type of cookie.
- The researchers are interested in how the visual and olfactory perception affects taste.
- This system also allows for interesting experiments, such as the participant “seeing” one type of cookie while “smelling” a different type. Note that the tubing is kept out of the view of the participant.
- Photo courtesy of Alan B. Craig

AR Software

1. Software involved directly in the AR application
2. Software used to create the AR application
3. Software used to create the content for the AR application
4. Other Software related to AR application

AR Software Components

1. Low-level programming libraries (e.g., tracking software)
2. Rendering and application building libraries
3. Plug-in software for existing applications
4. Standalone applications (e.g., content building, complete AR authoring)
5. Software to create the content for the AR application

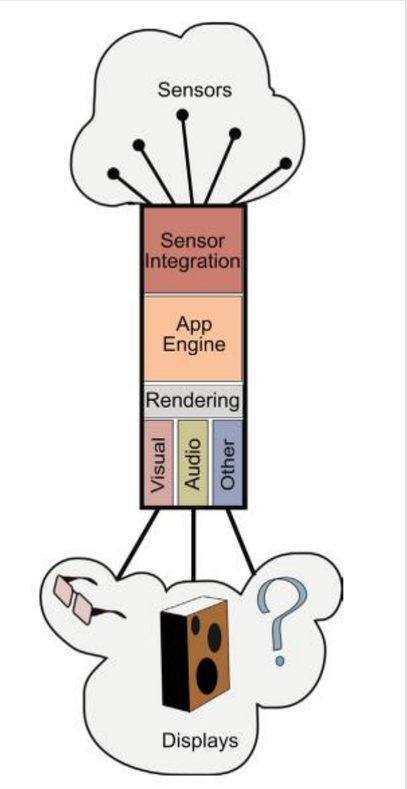
The key Component of a Game

1. Tracking library (to recognize the fiducial markers or features)
2. Model loading and animation (to provide the AR elements of the game)
3. Game logic/game engine (i.e., something that provides the actual “game” of the application)
4. Rendering software to provide the images (visual, audio, etc.) to the display(s). Note that many game engines include rendering software as part of the game engine

Software involved directly in the AR application

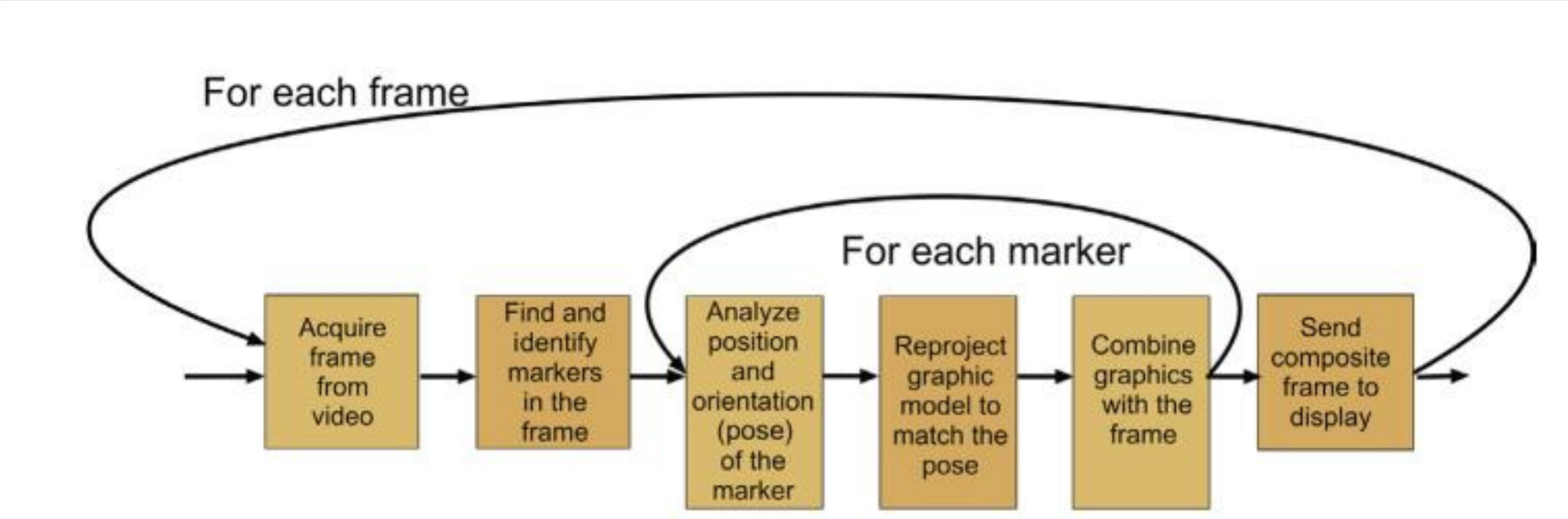
- Environmental acquisition (sensors)
- Sensor integration
- Application engine
- Rendering software (visual, audio, etc.)

This diagram shows the relationship between the core components (hardware and software) of an augmented reality system. The state of the environment is assessed by the sensors—the results of those are combined and fed to the main AR application. Based on the state of the total environment, the system renders the information as signals that are then fed to the various displays. This process repeats constantly, and the displays are updated anytime any piece of information changes. This might include the participant changing his or her position slightly or some other aspect of the environment or AR application changing



Software involved directly in the AR application

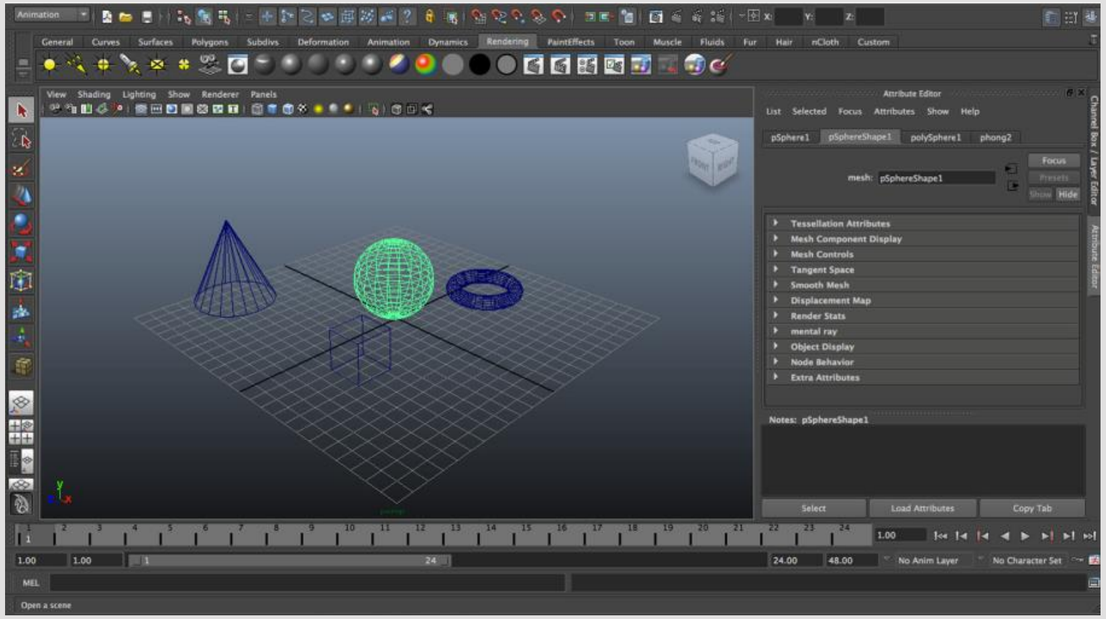
- Application engine

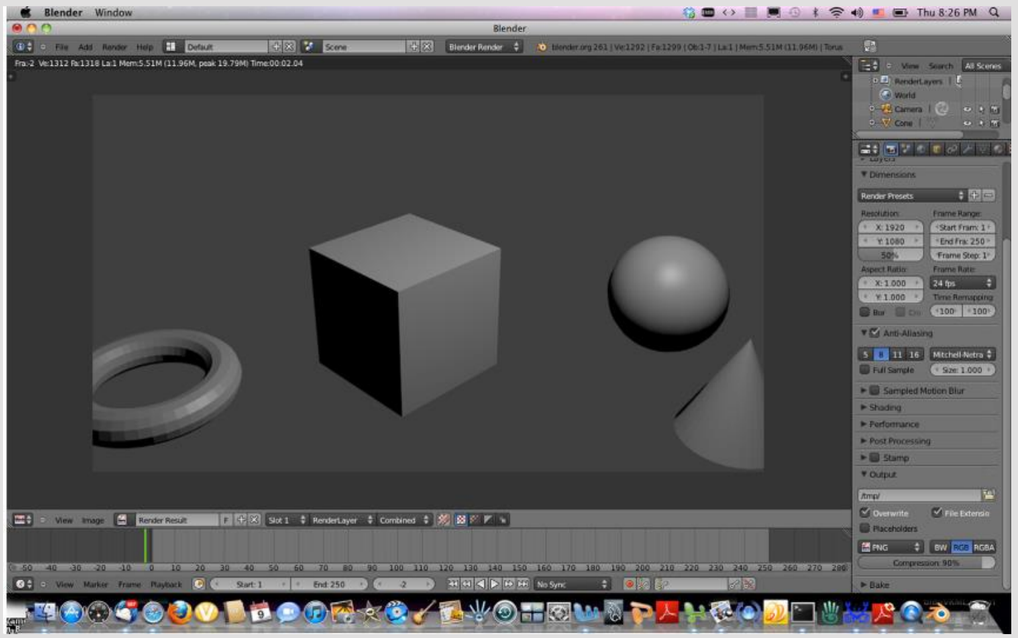


This diagram illustrates the basic flow of an augmented reality application. Diagram courtesy of Robert E. McGrath and Beverly Carver

Software used to create the AR application

- Software for Creating and Editing Three-Dimensional Graphics

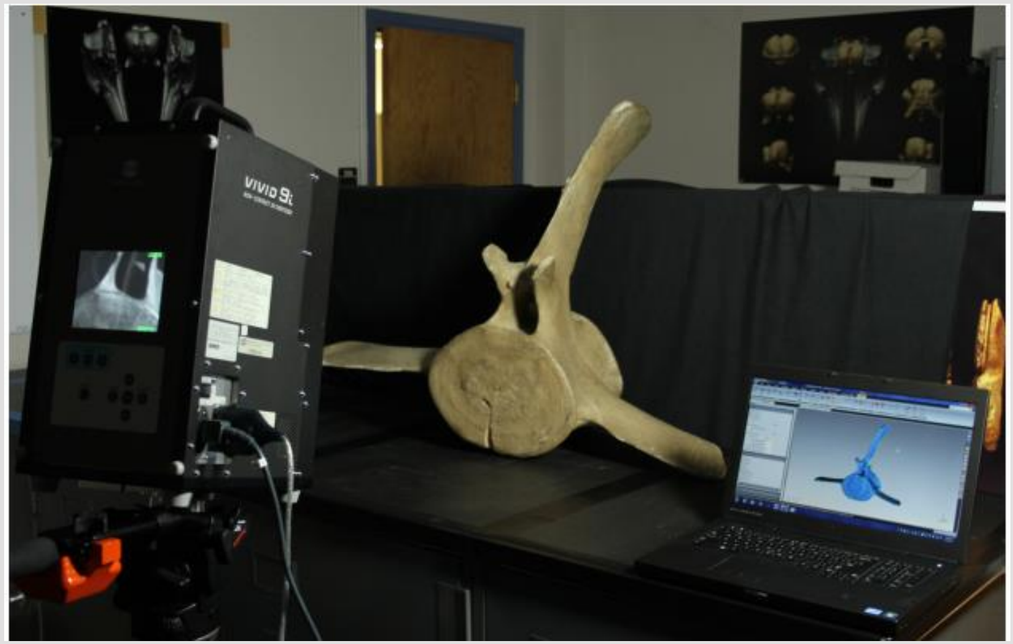




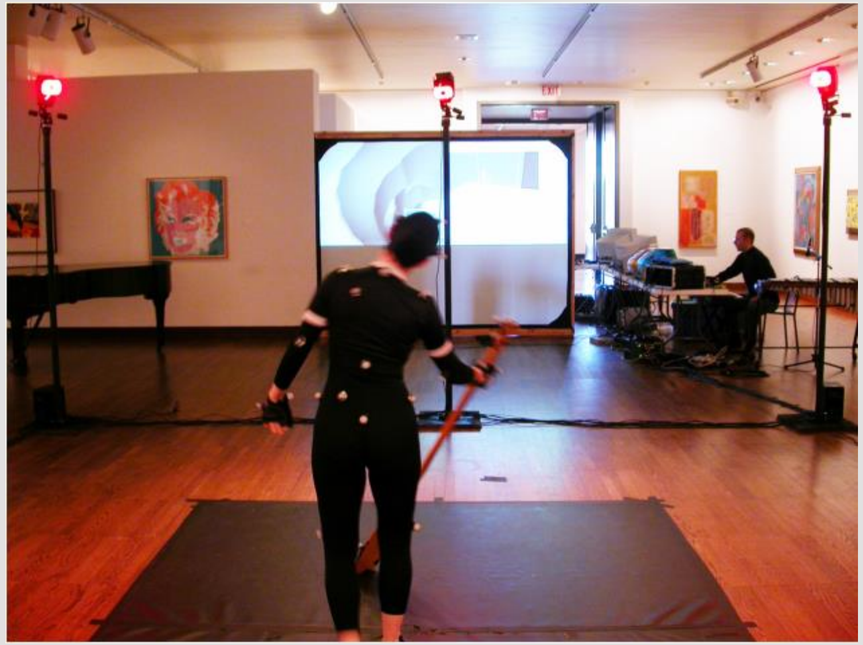
Autodesk Maya

Software used to create the AR application

- Software for Creating and Editing Three-Dimensional Graphics



3D scanner (Konica Minolta Vivid 9i



Motion capture systems use optical tracking to capture the motions of multiple objects. In this photo, there are a set of cameras and lights surrounding the area in which the tracking is done. The dancer is wearing small retroreflective spheres at each point their body is being tracked. In this case, the dancer's motions are controlling the computer graphic

AR Game Engine

- Unity, Vuforia Engine, EasyAR, Wikitude, HTML5, ARKit, ARToolkit















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Thank you

